

Homework 6.2.1

$$E[T_i] = \frac{1}{\lambda} ; \lambda = \frac{1}{E[T_i]}$$

$$\lambda = \frac{1}{\mu} \Rightarrow \frac{1}{0.4} = 2.5$$

(a) Let $X(t) = n$: n customers in the system

Transition

from n to $n+1$: arrival rate $\lambda = 2.5$

from n to $n-1$: departure rate $\mu \cdot \min(n, 3)$

$$\rightarrow q_{n,n+1} = 2.5$$

$$\rightarrow q_{n,n-1} = 0.4 \cdot \min(n, 3)$$

$$\rightarrow q_{n,n} = -(q_{n,n+1} + q_{n,n-1})$$

$$(b) \text{Embedd. } \Delta TMC = P_{22} = \frac{\lambda_2}{\lambda_2 + \mu} = \frac{2.5}{2.5 + 0.8} = \frac{25}{33} = 0.7576$$

$$(c) P_{22} = \frac{\mu}{\lambda + \mu} = \frac{1.2}{2.5 + 1.2} = \frac{1.2}{3.7} = \frac{12}{37} = 0.324$$

$$(d) \begin{cases} P_{i,i+1}(h) = \lambda h + o(h) \\ P_{i,i-1}(h) = \mu \cdot \min(i, 3) \cdot h + o(h) \\ P_{i,i}(h) = 1 - (\lambda + \mu \cdot \min(i, 3))h + o(h) \\ P_{ij}(h) = o(h) \text{ for } |i-j| > 1 \end{cases}$$

$$(e) P_{22} = \lambda h + o(h) \Rightarrow \lambda (0.03) \Rightarrow 3\lambda (0.03) \Rightarrow 3(0.4)(0.03) = 0.036$$

$$(f) P_{22} = 1 - (\lambda + \mu)h + o(h) \Rightarrow 1 - (2.5 + 0.8) \cdot 0.02 \Rightarrow 1 - 3.3(0.02) = 1 - 0.066 = 0.934$$

$$(g) P_{55} = 1 - \frac{(2.5 + 3(0.4)) \cdot 0.3}{1 - (1.4)} + o(h)$$

$$(h) P_{55} = 1 - 3.7(0.3) \Rightarrow 1 - 1.11 \Rightarrow \text{Negative}$$

exponential decay

$$P_{55} = e^{-3.7(0.3)} = e^{-1.11} \approx 0.329$$

Homework 6.2.2

$$\text{Birth rate} = \frac{2}{100} = 0.02 \Rightarrow \beta$$

$$\text{Death rate} = \frac{1}{100} = 0.01 \Rightarrow \delta$$

$$\text{Immigration rate} = 0.015 \Rightarrow \mu$$

(a) CTMC model

States: $X(t) = n$

Transition:

$n \rightarrow n+1$: due to birth or immigration

$$\text{Total rate: } q_{n,n+1} = n(\beta + \mu) = n(0.02 + 0.015) = 0.035n$$

$n \rightarrow n-1$: due to death

$$\text{Rate: } q_{n,n-1} = n \cdot \delta = 0.01n$$

$$q_{n,n+1} = 0.035n$$

$$q_{n,n-1} = 0.01n$$

$$q_{n,n} = -(q_{n,n+1} + q_{n,n-1}) = -0.045n$$

(b) Population is 30

$$q_{30,31} = 0.035(30) = 1.05$$

$$P(T > 0.03) = e^{-1.05(0.03)} = e^{-0.0315} = 0.969$$

(c) $P(X < 0.02)$?

rate of death

$$q_{0,22} = 0.01(30) = 0.3$$

$$P(T < 0.02) = 1 - e^{-0.3(0.02)} = 1 - e^{-0.006} = 0.006$$

$$(d) \begin{cases} q_{n,n+1} \cdot h + o(h) = 0.035n(h) + o(h) \\ 0.01n(h) + o(h) \\ 1 - (0.035n + 0.01n)h + o(h) = 1 - 0.045n \cdot h + o(h) \\ o(h) = |n-1| \end{cases}$$

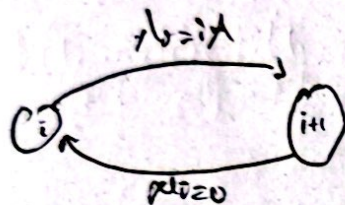
Homework 6.2.3

$\{X_t, t \geq 0\}$ - CTMC

$X(t)$ = no. of cars at time t
 $t = 0, 1, 2, 3, \dots$

Arrivals occur at the rate of λ per person per unit time

No. of cars $M \geq 0$



Transition Probability

$$P_{i,i+1}(h) = \lambda h + o(h)$$

$$P_{i,i}(h) = 1 - \lambda h + o(h)$$

$$P_{i,j}(h) = o(h), \quad j \neq i, i+1$$

Homework 6.3.1

$$P_{0,0}(t) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{\lambda + \mu} e^{-(\lambda + \mu)t}$$

$$P_{1,1}(t) = \frac{\lambda}{\lambda + \mu} + \frac{\mu}{\lambda + \mu} e^{-(\lambda + \mu)t}$$

$$P_{0,1}(t) = 1 - P_{0,0}(t), \quad P_{1,0}(t) = 1 - P_{1,1}(t)$$

Let's

$$P_{0,0}(t) = P(X(t) = 0 | X(0) = 0)$$

$$P_{0,1}(t) = 1 - P_{0,0}(t)$$

$$\frac{dP_{0,0}(t)}{dt} = -\lambda P_{0,0}(t) + \mu(1 - P_{0,0}(t)) = -(\lambda + \mu)P_{0,0}(t) + \mu$$

By CDE $\frac{dP_{0,0}}{dt} + (\lambda + \mu)P_{0,0} = \mu$

$$GS: P_{0,0}(t) = C e^{-(\lambda + \mu)t} + \frac{\mu}{\lambda + \mu}$$

$$Case: P_{0,0}(0) = 1$$

$$1 = C + \frac{\mu}{\lambda + \mu} \Rightarrow C = \frac{\lambda}{\lambda + \mu}$$

$$Therefore, P_{0,0}(t) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{\lambda + \mu} e^{-(\lambda + \mu)t}$$

$$P_{0,1}(t) = 1 - P_{0,0}(t) = \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)t})$$

By symmetry

$$P_{1,1}(t) = \frac{\lambda}{\lambda + \mu} + \frac{\mu}{\lambda + \mu} e^{-(\lambda + \mu)t}, \quad P_{1,0}(t) = 1 - P_{1,1}(t)$$

$$(1) P(X(5) = 1 | X(0) = 0) = P_{0,1}(5) = \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)5})$$

$$(2) P(X(5) = 0 | X(0) \geq 0) = P_{0,0}(5) = P_{0,0}(t) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{\lambda + \mu} e^{-(\lambda + \mu)5}$$